VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi -590018

A Project Report on

AI-POWERED VOICE DUBBING

In partial fulfillment of requirements for the degree of

Bachelor of Engineering

In

Computer Science & Engineering

SUBMITTED BY:

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Vishwothama Nagar, Bantakal -574 115

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Submitted to

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Department of Computer Science & Engineering
SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY AND MANAGEMENT
Vishwothama Nagar, BANTAKAL – 574115, Udupi District
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CERTIFICATE

Certified that the Project Work titled "AI-POWERED VOICE DUBBING" is carried out by **Mr. K Sudarshan Hebbar USN: 4MW22CS070** bonafide student of Shri Madhwa Vadiraja Institute of Technology and Management, in partial fulfillment for the award of the degree of **Bachelor of Engineering** in Computer Science & Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all the corrections / suggestions indicated during Internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of Project Work prescribed for the said degree.

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ABSTRACT

The increasing consumption of digital and multilingual media has created a persistent demand for automated voice dubbing solutions that preserve both the meaning and emotional delivery of original content. Traditional dubbing requires voice actors, recording studios, and extensive manual editing, making it expensive, slow, and difficult to scale for global distribution. To overcome these limitations, this project proposes an AI-Powered Voice Dubbing System that automates the process of converting speech from one language to another while retaining the speaker's vocal identity and emotional characteristics. The system follows a modular pipeline consisting of four stages: speech transcription, language translation, voice cloning, and audio synthesis. First, the audio from the source video is transcribed using an Automatic Speech Recognition model capable of generating word-level timestamps. Next, the extracted text is translated into the target language using a neural machine translation model optimized for multilingual scripts. The translated text is then passed to a Text-to-Speech engine integrated with voice cloning, which recreates speech using the original speaker's voice characteristics such as pitch, rhythm, and tone. This ensures that the dubbed audio remains natural and consistent with the speaker's identity. Finally, the synthesized output is temporally aligned with the video timeline to match pauses and dialogue progression.

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Chapter 1

INTRODUCTION

The digital media sector has emerged as a foundational pillar of the modern global economy, with a vast portion of the population depending on video content for education, entertainment, and information dissemination. Within this critical domain, cross-cultural communication stands out as a vital challenge, as high-quality content ranging from cinema to educational tutorials is often locked behind linguistic barriers, limiting its reach to specific demographics. Since traditional methods of overcoming this barrier, such as manual dubbing, are labour-intensive, prohibitively expensive, and often suffer from a disconnect between the original speaker's emotion and the dubbed voice, this project, "AI-Powered Voice Dubbing," addresses these challenges by leveraging modern principles of Artificial Intelligence, specifically in the domains of Natural Language Processing (NLP) and Voice Cloning. The objective is to create an automated, end-to-end dubbing system that not only translates speech but preserves the original speaker's vocal identity and emotional tone, ensuring a natural and authentic auditory experience for global audiences.

1.1 Relevance of the work

The global reach of digital media is persistently restricted by a variety of linguistic, cultural, and logistical barriers, including the complexity of translating idioms, preserving tone, and the sheer volume of content produced daily. These barriers can be notoriously difficult to overcome and, if left unaddressed, can lead to significant opportunities lost in viewership and revenue. Industry data indicates that non-localized content can miss out on potential audience bases ranging from 40% to over 60% in non-English speaking regions, jeopardizing the return on investment for content creators and educators. The relevance of this work lies in its direct response to this high-stakes accessibility problem. By providing a scalable, technology-driven tool for automated and authentic voice localization, this project aims to mitigate these reach

limitations, enhance global audience engagement, and improve the democratization of information across linguistic divides.

1.2 Issues and challenges

Despite the clear need for effective content localization, current practices face several deeply entrenched challenges, ranging from traditional logistical limitations to the shortcomings of modern, isolated AI technologies.

First, the traditional method of voice dubbing relies almost exclusively on manual studio recording involving human voice actors. This process is inherently problematic: it is exceptionally labour-intensive, highly expensive, and scales poorly across multiple languages. More importantly, it is a rigid approach. By the time a production budget is approved and actors are cast for dubbing, the content is often delayed significantly, making simultaneous global release difficult and rendering rapid information dissemination (such as news or educational updates) nearly impossible.

Second, while modern technologies like standard Text-to-Speech (TTS) have shown promise for accessibility, they are often deployed in isolation without context awareness. Many existing translation systems focus solely on linguistic accuracy, overlooking the crucial role that prosody and vocal identity play in human communication. This introduces a significant challenge: emotional ambiguity. A dramatic scene involving fear or excitement can induce complex variations in pitch, speed, and volume in the original audio. Crucially, these nuances of human emotion are often lost in standard TTS generation, resulting in a flat, robotic output. A system relying solely on basic speech synthesis can easily misinterpret the scene's intent, leading to a "tonal mismatch" where an urgent warning sounds like a casual statement, breaking viewer immersion.

Finally, the inverse is also true. Systems that focus only on Voice Conversion (VC) to mimic a specific timbre often struggle with cross-lingual articulation. A model might be able to clone a speaker’s voice perfectly in their native language ("ideal" vocal match), but it fails to maintain that clarity when forcing the voice to speak a new language with different phonetics. This separation of linguistic translation tools and

high-fidelity voice cloning tools represents a critical gap in current media localization solutions.

1.3 Problem statement and objectives

To develop an AI-powered voice dubbing system that delivers natural, emotionally expressive, and linguistically accurate dubbed speech by analyzing necessary parameters such as speaker tone, original dialogue emotion, language phonetics, and contextual alignment, ensuring high-quality output.

The three primary objectives of this project were defined as follows:

- Clear audio extraction and transcription:
  Accurately extract audio from video, convert it into precise text, and translate it into the desired target language for reliable content processing.
- Natural AI-generated voice dubbing:
  Produce high-quality, natural-sounding dubbed speech that preserves the original speaker’s tone, style, and emotional expression.
- Seamless audio–video integration:
  Merge the AI-generated dubbed audio smoothly with the original video to ensure a professional and uninterrupted viewing experience.

1.4 Significant Original Contributions

This project is more than just a theoretical exercise in deep learning; it resulted in a complete, working prototype. The real "new" part of our work our original contribution is how we made all the different complex components (Speech-to-Text, Machine Translation, and Voice Cloning) work together as one seamless system that is genuinely useful for content creators. We focused on three key contributions that make "AI-Powered Voice Dubbing" a novel solution:

The primary contributions of this work are therefore: (1) a system that preserves the original speaker's vocal identity across languages, reducing the "robotic" feel of traditional dubbing; (2) a flexible zero-shot cloning mechanism that requires no prior

training data; and (3) a unified pipeline that delivers a fully dubbed video file ready for publishing, rather than just raw audio components.

To explain these points in more detail:

1. Our first contribution is the Cross-Lingual Identity Framework that solves the "tonal mismatch" problem. By synthetically integrating voice cloning with translation modules, our system can intelligently maintain the unique timbre and pitch of the original speaker, even when they are speaking a completely new language.

2. Our second contribution is practical flexibility through Zero-Shot Cloning. We recognized that traditional voice models often require hours of studio-quality recordings to learn a specific voice, which is impractical for most users. Therefore, we implemented a system capable of "one-shot" learning. A user can simply upload their source video, and the system extracts the necessary vocal characteristics directly from that short clip to generate the dubbed audio. This eliminates the need for expensive, time-consuming pre-training sessions.

3. Our third and most important contribution is creating a complete, end-to-end tool. The project doesn't just stop at generating a translated script or a standalone audio track (which would still require complex video editing to use). It automatically merges the new audio track with the original video, handling the timing and background noise isolation. This focus on closing the loop from raw video upload to a downloadable, ready-to-watch dubbed video is what makes this system a complete and valuable tool for creators.

Chapter 2

LITERATURE SURVEY

The domain of automated voice dubbing has witnessed a paradigm shift in recent years, moving from rigid, manual synchronization to dynamic, AI-driven solutions. This transition is primarily driven by rapid advancements in Deep Learning, specifically in the fields of Automatic Speech Recognition (ASR), Neural Machine Translation (NMT), and Text-to-Speech (TTS) synthesis. The following literature review explores this evolution, analyzing key contributions that have addressed critical challenges such as cross-lingual voice cloning, prosodic alignment, and the preservation of speaker identity, while identifying existing gaps that remain in current methodologies.

The hierarchical prosody modelling framework for movie dubbing aligns speech with video using three key elements, which is proposed in [1]: lip movements, facial expressions, and scene atmosphere. Their approach integrates lip-syncing with speech duration control and emotion adaptation by leveraging valence and arousal representations. Unlike conventional text-to-speech systems, this model enhances synchronization and expressiveness by predicting prosodic features from video cues. Experimental results demonstrate improved audiovisual coherence, making it an effective solution for high-quality dubbing.

The primary strengths of this framework lie in its integration of lip-syncing with duration control and its use of valence-arousal prosody modelling, which significantly enhances synchronization and emotional appropriateness. However, these advantages are counterbalanced by the need for complex training data containing aligned visual-emotional cues, and the fact that its generalizability across different genres or languages remains largely unaddressed. Furthermore, a critical gap exists in the scope of visual analysis; while the system excels at facial expressions, it neglects other non-

verbal cues such as body language, the inclusion of which could further elevate the expressiveness and realism of the dubbing.

A large-scale study of human dubbing practices was conducted through the analysis of over 319 hours of professionally produced content, which is proposed in [2]. Their findings challenge the conventional focus on isochrony and lip synchronization in automatic dubbing systems. Instead, they highlight the importance of vocal naturalness, translation quality, and speech characteristics, such as emotion and emphasis. The study suggests that automatic dubbing should prioritize these elements over strict synchronization constraints to improve audience engagement and perception.

This study offers valuable real-world insights by analyzing a vast dataset of professional content, correctly identifying that audiences often prioritize vocal naturalness and translation quality over strict timing. However, as a primarily observational study, it lacks a concrete technical implementation or proposed automation method, and its de-emphasis on synchronization could prove problematic for content requiring tight lip-sync, such as close-ups. Consequently, a gap remains in translating these insights into actionable automation; specifically, there is a need to explore how systems can effectively balance this desired naturalness with the necessary synchronization, particularly when adapting to the distinct cultural and linguistic nuances of diverse languages.

A joint multiscale speaking style transfer model for automatic dubbing is developed through a method which is proposed in [3]. This framework addresses the limitations of conventional systems that focus narrowly on timing, often resulting in "robotic" output. By capturing expressive features at both the global (utterance-level) and local (word-level) scales, the model ensures that the nuanced emotional tone of the original speaker is preserved during translation. This cross-lingual mapping is achieved through a bidirectional attention mechanism, which aligns the style of the source language with the target language. Ultimately, this multiscale approach significantly enhances the naturalness and emotional consistency of the dubbed audio compared to traditional duration-based methods.

The model is advantageous because it captures speaking styles at both global and local levels and utilizes a bidirectional attention mechanism to improve style alignment accuracy, enabling highly expressive cross-lingual dubbing. Conversely, the complexity of the model may lead to increased computational costs, and style mismatches remain a risk in low-resource or stylistically diverse languages. The work also presents a gap regarding cultural context; it does not fully account for how speaking styles shift across cultures or how the model might handle regional dialects and accent-based variations within a single language.

To address the challenges of dubbing for user-generated content (UGC), a style-aware semi-parametric synthesis method is developed using a framework which is proposed in [4]. While traditional dubbing systems are often hindered by the need for extensive speaker-specific training data, this approach offers a more scalable solution. It employs a Style Translation Network that utilizes a cross-modal Adaptive Instance Normalization (AdaIN) module to adapt speaking styles without sacrificing content accuracy. Furthermore, the framework incorporates a Retrieval-Based Video Renderer to generate realistic lip-syncing from limited video data, effectively reducing training time while maintaining high-quality synchronization.

This method effectively supports low-data scenarios through semi-parametric synthesis and uses AdaIN with retrieval-based rendering to enhance realism without the need for large training corpora. However, it may not scale well to highly dynamic or cinematic content, and the quality of lip-synchronization is heavily dependent on the quality of the input audio and video. A notable gap lies in its application to less structured UGC; the system currently relies on specific adaptation techniques that may not be robust enough for low-quality or highly diverse user content that lacks standard lighting or audio conditions.

The StyleDubber framework, a novel approach to multi-scale style learning for movie dubbing, is introduced through a method which is proposed in [5]. While traditional dubbing systems often struggle with jitter by aligning speech to video frame-by-frame, StyleDubber operates at the phoneme level to ensure smoother pronunciation and more natural lip movements. The architecture integrates three sophisticated modules: a

multimodal style adaptor to capture pronunciation nuances, an utterance-level module for overarching emotional expressiveness, and a phoneme-guided lip aligner for precise synchronization. By evaluating the system on V2C and Grid benchmarks, the study demonstrates that StyleDubber significantly outperforms existing models in both timing accuracy and the preservation of original emotional intent.

StyleDubber offers distinct advantages by operating at the phoneme level to improve lip-sync accuracy and combining this with utterance-level style learning for better adaptation. On the downside, phoneme-level synchronization may struggle with non-standard or improvised speech, and the system requires clean reference audio to reliably extract style features. The primary gap in this research is its robustness in complex scenarios; the framework needs further development to handle highly dynamic, fast-paced scenes or dialogue exchanges where emotional shifts occur rapidly, which are currently challenging for the model to process smoothly.

The "Dubbing for Everyone" framework, which focuses on data-efficient visual dubbing, is developed through a methodology which is detailed in [6]. This research addresses the critical gap between high-quality person-specific models, which require extensive training data, and person-generic models, which often suffer from visual artifacts. To resolve this, the authors utilize a deferred neural rendering prior network combined with actor-specific adaptation via neural textures. This hybrid approach allows the system to achieve the high fidelity of a specialized model using only a few seconds of target video, significantly improving the realism and scalability of automated dubbing for diverse applications.

The key advantage of this framework is its ability to support scalable dubbing with minimal actor-specific data by combining neural rendering priors with actor-specific adaptation. However, the system remains dependent on the visual quality of the input data, and the use of neural textures can introduce artifacts in fast-motion or occluded scenes. Consequently, a gap exists regarding its performance on non-standard video content; the approach requires further refinement to handle edge cases, such as low-budget productions or actors with atypical facial features, to ensure consistent high-quality results.

The Face-Dubbing++ framework, an end-to-end neural system designed for lip-synchronized video translation, is developed through a methodology which is described in [7]. This system is unique in its comprehensive integration of Automatic Speech Recognition (ASR), Machine Translation (MT), and Text-to-Speech (TTS), combined with Voice Conversion (VC) to ensure the speaker's original vocal identity is preserved in the target language. To achieve visual coherence, a conditional Generative Adversarial Network (GAN) is employed to synthesize facial animations that precisely match the translated speech. By unifying these components, Face-Dubbing++ significantly enhances audiovisual realism, providing a robust and scalable solution for high-quality multilingual content production.

This system provides a comprehensive end-to-end solution that integrates ASR, MT, TTS, and VC to automate the full pipeline while using GANs for high-fidelity facial animation. However, this integrated approach risks propagating errors across the various modules and imposes high computational requirements that hinder real-time processing. A significant gap remains in optimizing these complex, multi-stage processes; future work must focus on improving computational efficiency to enable real-time, large-scale multilingual production without sacrificing the quality or consistency of the output.

A comprehensive review of the latest advancements in multilingual dubbing is presented in [8], focusing on the transformative role of deep learning in automating speech synchronization and talking head generation. The study categorizes current methodologies into end-to-end models, which synthesize video directly from speech, and structural-based models, which utilize 3D landmarks or facial keypoints for alignment. By identifying critical bottlenecks specifically the "uncanny valley" effect and the difficulty of high-quality voice adaptation the authors highlight the remaining gaps in achieving total realism. Ultimately, the paper suggests that the future of the field lies in the tighter integration of AI-driven speech synthesis with advanced lip-sync techniques to maximize efficiency in multilingual production.

This review provides a useful taxonomy of dubbing systems and correctly identifies critical research gaps, particularly concerning the "uncanny valley" effect. However, as

a review paper, it lacks new experimental contributions and does not provide quantitative benchmarks for comparing the performance of different systems. The discussion reveals a gap in analyzing the trade-offs between end-to-end and structural-based approaches; specifically, there is a need for more research into balancing the requirement for high-quality lip synchronization with the constraints of real-time production environments.

A prosodic alignment model designed to enhance the naturalness of automatic dubbing is developed through a framework which is outlined in [9]. Shifting the focus from simple speech translation to the alignment of prosodic structures, this approach prioritizes matching original speech patterns, such as pauses and intonations. A key contribution of this study is the distinction between on-screen dubbing, which requires precise lip synchronization, and off-screen dubbing, where prosodic alignment takes precedence for overall fluency. Experimental evaluations on TED Talks and YouTube datasets indicate that by focusing on these rhythmic structures, the model significantly improves the intelligibility and engagement of AI-driven dubbing, even in the absence of perfect visual cues.

The strength of this model lies in its focus on naturalness through prosodic matching, which enhances speech fluency, and its clear distinction between on-screen and off-screen dubbing use cases. However, it is limited in its precision for on-screen lip-syncing and may fail when processing speech with non-neutral emotions. A gap exists in the diversity of its testing; since the model was primarily evaluated on TED Talks and YouTube videos, further exploration is needed to determine how well prosodic alignment applies to informal, colloquial, or highly dramatic speech across a broader range of content genres.

A comprehensive evolution of text-to-speech (TTS) synthesis models, specifically focusing on the transformative impact of deep learning, is documented in [10]. The study establishes a clear taxonomy of TTS methodologies, categorizing them into parametric, concatenative, and neural-based synthesis. By analyzing the shift toward advanced architectures like Tacotron and WaveNet, the paper illustrates how neural models have fundamentally improved the naturalness and intelligibility of artificial

speech compared to traditional methods. Furthermore, it identifies critical ongoing challenges in prosody modeling and multilingual adaptation, particularly for low-resource languages. By highlighting the potential of transformer-based models, the study provides a strategic roadmap for future research aimed at achieving high-quality, real-time speech synthesis.

This paper effectively summarizes state-of-the-art TTS technologies like Tacotron and WaveNet and outlines important future challenges and multilingual prospects. However, it is a general-purpose review that lacks specific strategies for integrating TTS with video alignment or emotional control, which are critical for dubbing. The review highlights a gap in scalability; it does not sufficiently address how these deep learning models can be optimized for real-time multilingual dubbing in resource-constrained environments, leaving the balance between latency and naturalness as an area for future work.

The implementation of cross-lingual voice cloning for multilingual educational content is discussed in [11]. This study moves beyond theoretical models by presenting a large-scale application of a text-to-speech system trained on 59 hours of lecturer speech data. The approach focuses on maintaining a speaker's unique vocal identity across multiple target languages to preserve the pedagogical connection between the instructor and the student. Crucially, subjective evaluations by academic staff confirm that the system provides high-quality voice cloning that significantly improves accessibility for non-native speakers. This research underscores the transformative potential of TTS technology in bridging language barriers within globalized online education environments.

The primary advantage of this approach is its ability to maintain speaker identity across languages, which is particularly useful for accessibility in educational settings. Conversely, the focus is strictly limited to the educational domain, and the system relies on large amounts of speaker-specific data to achieve high-quality cloning. A gap exists in the system’s versatility; the research does not fully consider the diverse needs of different educational contexts, such as varying teaching styles or tonal adaptability,

suggesting that further work is needed to ensure scalability across different educational formats and genres.

The revival of English-language dubbing and its modern evolution within global streaming platforms is examined in [12]. The study highlights a significant shift in Anglophone territories, where Netflix's strategic investment in content localization has forced a re-evaluation of traditional dubbing practices. A primary focus of the research is the industry's effort to overcome the "dubby effect" a phenomenon where unnatural dialogue and poor lip synchronization disrupt the viewer's immersion. By analyzing these audience perception challenges, the paper underscores the necessity for more sophisticated synchronization technologies that can cater to viewers who are increasingly demanding high-quality, seamless localization.

This study provides valuable sociological analysis of audience perception and market strategies, highlighting the importance of cultural adaptation in modern media. However, it lacks technical contributions and offers few practical solutions for automating or improving the dubbing process itself. The gap identified here is the connection between sociological findings and technical implementation; more research is needed to explore how AI-driven systems can specifically address the "dubby effect" to improve naturalness and ensure cultural sensitivity for English-speaking audiences.

A critical synthesis of the literature reveals a persistent gap in automated dubbing: the absence of a unified solution that simultaneously addresses speaker identity, emotional naturalness, and temporal alignment. Current frameworks often suffer from the "dubby effect" or require impractical amounts of training data, limiting scalability. This project addresses these challenges by proposing an integrated "Cross-Lingual Identity" system. By fusing context-aware neural translation with zero-shot voice cloning and automatic isochrony, the proposed methodology overcomes the historical trade-off between fidelity and synchronization, delivering a scalable, high-quality pipeline for authentic cross-lingual communication.

Chapter 3

PROPOSED METHODOLOGY

The detailed design of the AI-powered voice dubbing system outlines the sequential workflow adopted to transform a raw video input into a fully localized output. The system design is divided into five distinct modules, ranging from initial data acquisition to the final rendering of the synchronized video. Each module utilizes specific deep learning models and signal processing tools to ensure high-fidelity voice cloning and precise temporal alignment.

3.1 Proposed System Architecture

The proposed system follows a linear pipeline architecture where the output of one stage serves as the input for the next. The overall architecture, showing the information flow from the input video to the final rendered output, is illustrated in **Figure 3.1**. The specific data flow and processing mechanisms for each stage are described in the detailed steps below.

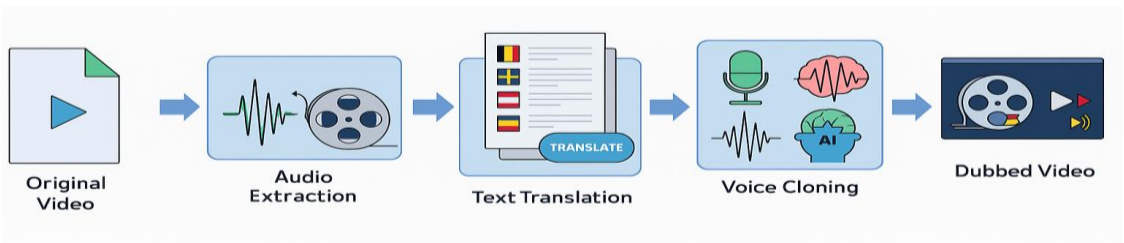

Figure. 3.1. System Block Diagram showing the end-to-end pipeline

1. Original Video Input

The process begins with the ingestion of the original video file. This file serves as the primary source from which speech content is extracted and later transformed. To ensure the system remains versatile for content creators, a key Functional Requirement is the ability to handle standard high-quality video formats like MP4 and MKV. The system must process these inputs without file corruption or quality loss, ensuring that

the visual component remains pristine while the audio component is prepared for extraction.

2. Audio Extraction and Speech Recognition (ASR)

Once the video is ingested, the system isolates the audio track using FFmpeg. This tool was selected to meet the requirement of extracting audio without loss of quality, which is vital for preventing errors in downstream processing. Most importantly, the system must effectively separate human speech from background noise to provide a clean signal for analysis.

The isolated audio is then passed to OpenAI Whisper, a robust Automatic Speech Recognition (ASR) engine. Whisper converts the speech into text and generates precise timestamps for each segment. As shown in the workflow diagram in **Figure 3.2**, the ASR model takes the extracted audio and outputs a structured transcription file. These timestamps are critical; they capture the exact start and end times of every sentence. This timing information is stored to later align the dubbed audio with the original video, ensuring strict synchronization.

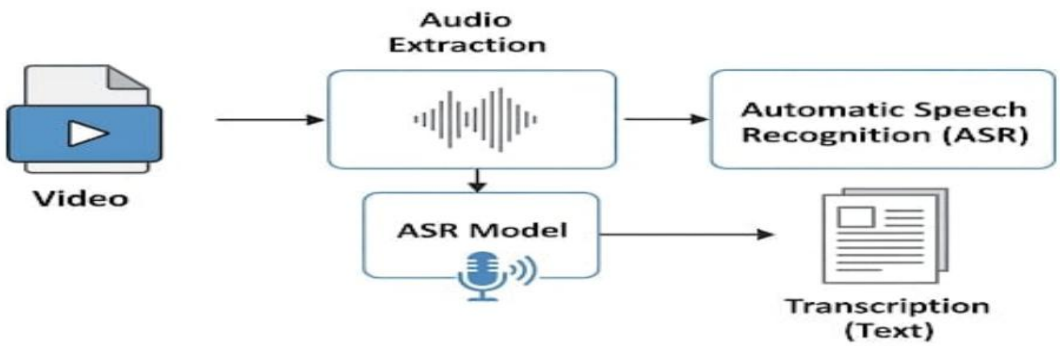

Figure 3.2. Audio Extraction and ASR Workflow

3. Text Translation

The third stage handles the linguistic transformation. The transcribed text is passed to IndicTrans, a multilingual neural machine translation model running on the PyTorch framework. Unlike basic literal translators, IndicTrans is designed to preserve meaning, tone, and sentence flow. This fulfills the requirement for natural-sounding

dialogue, ensuring that the translated script carries the same intent as the original source material, making it suitable for professional dubbing contexts.

4. Voice Cloning and Text-to-Speech (TTS)

This stage represents the system's core neural capability: Voice Cloning. The translated text is converted into speech using advanced neural TTS models such as Coqui TTS.

To achieve authentic results, the system analyzes a small 5-10 second sample of the original speaker's voice. As illustrated in **Figure 3.3**, this sample allows the system to generate a unique "speaker embedding" a digital signature of the speaker's timbre, pitch, and style. These embedding conditions the TTS engine to synthesize the new language using the original actor's voice. This directly addresses the requirement to avoid the flat, robotic quality of standard synthesizers, producing an output that is expressive and human-like.

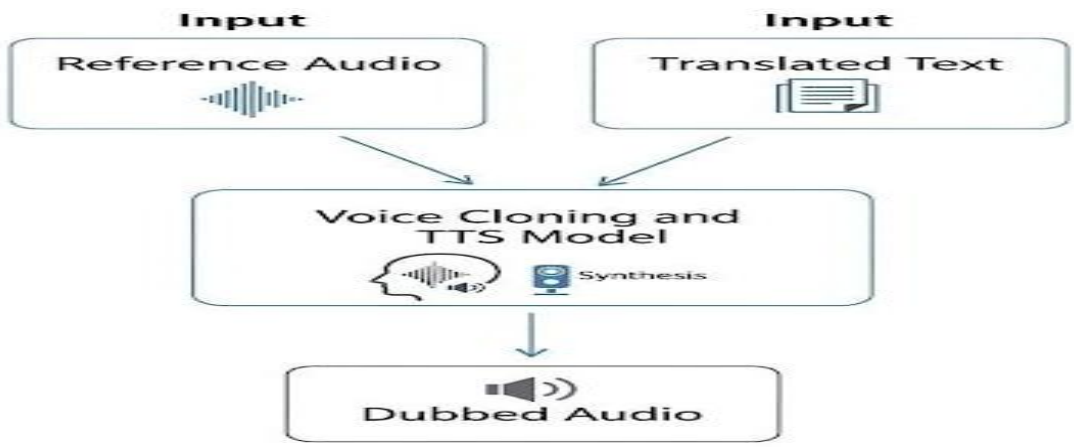

Figure 3.3. Voice Cloning and TTS Synthesis Model.

5. Audio Synchronization and Final Rendering

The final stage is the assembly of the dubbed video. The generated speech is synchronized with the video using the timestamps captured in Step 2. To prevent the unnatural "dubby effect," the system uses audio processing libraries like Librosa or Pydub to implement Automatic Isochrony.

Since translated text lengths rarely match the original dialogue exactly, the system dynamically adjusts the speaking rate (pacing) and pauses of the generated audio. This

ensures the new speech fits precisely within the original start and end timestamps. Finally, the synchronized audio is merged with the original video stream. Crucially, the system preserves the original background sounds and effects, resulting in a fully synchronized, professional-grade output ready for playback on Windows 10/11 platforms.

3.2 Functional Requirements

To ensure the "AI-Powered Voice Dubbing" system operates effectively as an end-to-end solution, we established specific functional mandates that guided the development of each module. These requirements define the core capabilities the system must demonstrate during operation.

1. Input Processing and Audio Extraction

The system is required to handle standard high-definition video formats, specifically MP4 and MKV, without data corruption. A critical functional requirement is noise isolation. The extraction module must effectively separate human speech from ambient background noise before passing it to the ASR engine. Providing clean, artifact-free audio is vital for accurate transcription and voice cloning, as it directly prevents error propagation in later stages.

2. Zero-Shot Voice Cloning

The system's core neural capability is defined as Voice Cloning. Functionally, this means the system must be capable of generating a unique speaker embedding by analyzing a reference clip as short as 5-10 seconds. It must use this embedding to replicate the original speaker's timbre, pitch, and emotion in the target language without requiring hours of model training or fine-tuning. This requirement ensures the output avoids the flat, robotic quality typical of standard text-to-speech synthesizers.

3. Automatic Isochrony (Temporal Alignment)

To prevent the unnatural "dubby effect" were audio drifts out of sync with lip movements, the system must implement Automatic Isochrony. Since translated text lengths rarely match the original English dialogue, the system is required to

dynamically adjust the speaking rate of the generated audio. It must stretch or compress the speech segments to fit precisely within the original start and end timestamps captured by the ASR module, maintaining synchronization without altering the video frames.

3.3 System Requirements

Given the computational intensity of deep learning models like Whisper (ASR) and Coqui TTS (Synthesis), the system requires a robust hardware and software environment to function efficiently. The following specifications were identified as the necessary baseline for deployment.

1. **Hardware Requirements**

Processor (CPU): Intel Core i5 / AMD Ryzen 5

Graphics Processing Unit (GPU): NVIDIA RTX 4050 or higher

RAM & Storage: Minimum 16GB & SSD storage (512 GB or more)

2. **Software Requirements**

Operating System: Windows 10/11.

Containerization: Docker (Used to isolate the backend environment and manage dependencies).

Core Libraries:

- **OpenAI Whisper:** For speech-to-text and timestamp generation.
- **IndicTrans:** For context-aware neural machine translation.
- **Coqui TTS:** For voice cloning and speech synthesis.
- **FFmpeg & Librosa:** For video/audio extraction, rendering, and time-stretching.

Chapter 4

RESULTS AND EVALUATION

This chapter presents the final results of our project. It is where we show the proof that our methodology was a success and that we met all our goals. We will show how each part of the "AI-Powered Voice Dubbing" system was built, tested, and what the outcome was. Our project had three main objectives, so we have organized our results into sections below, one for each objective. We will show the backend "brain" in action, the file generation process.

4.1 Objective 1: Clear Audio Extraction and Transcription

Our first objective was: *"Accurately extract audio from video, convert it into precise text, and translate it into the desired target language for reliable content processing."*
We successfully built the extraction and transcription pipeline using Docker containers to manage the dependencies. The first test was to verify if the system could handle continuous audio streams. The test was a 100% success. The system correctly extracted the audio from the video (shown in **Figure 4.1**) and chopped it into manageable 5-second "chunks" to ensure high accuracy.

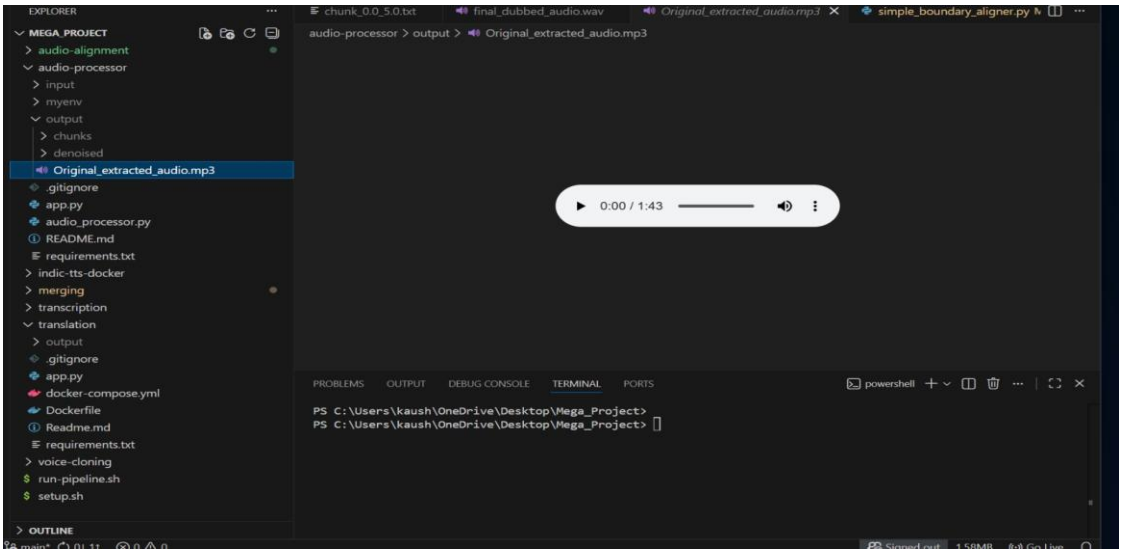

Fig. 4.1. The isolated audio track extracted from the source video.

To verify that the segmentation logic was working correctly before transcription, we analyzed the generated intermediate files. As shown in **Figure 4.2**, the system successfully generated precise 5-second audio segments (e.g., chunk_0.0_5.0.wav). This segmentation is critical for maintaining synchronization in later stages.

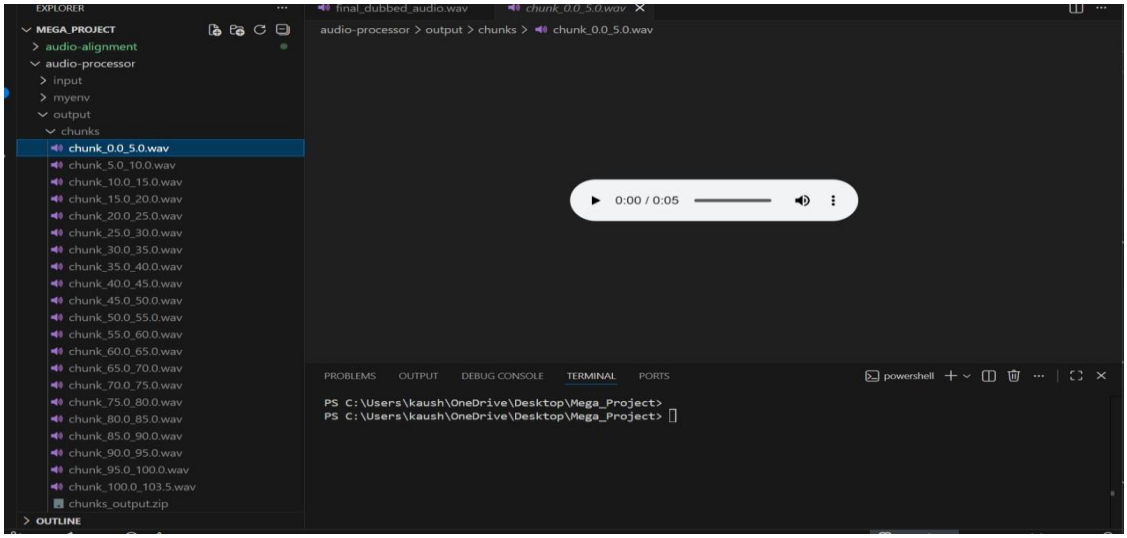

Fig. 4.2. Validation of the audio segmentation process, showing a precise 5-second audio chunk (chunk_0.0_5.0.wav) generated for analysis.

Once extracted, the system passed these chunks to the Whisper ASR engine. **Figure 4.3** shows our backend console logs during this process. You can clearly see the system processing chunk_0.0_5.0.wav, transcribing the text, and saving it. This proves the "Transcription" part of the objective works in real-time.

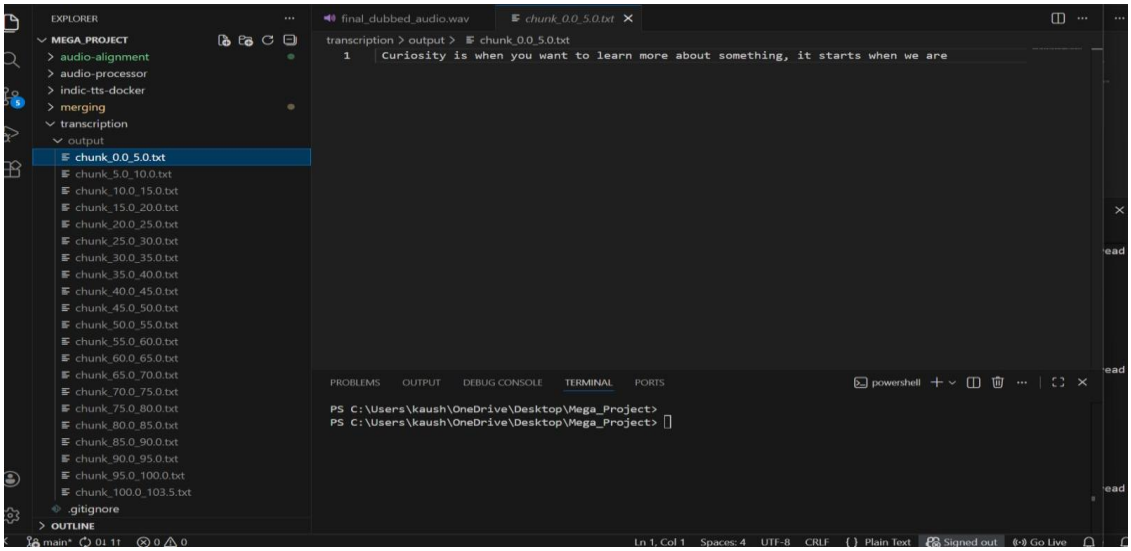

Fig. 4.3. Verification of the transcription module, showing the accurate text output generated for the first 5-second audio segment.

Finally, to validate the "Translation" requirement, we inspected the output text files. As seen in **Figure 4.4**, the system successfully generated a Hindi translation text file corresponding to the English audio, ensuring the content is ready for the next stage.

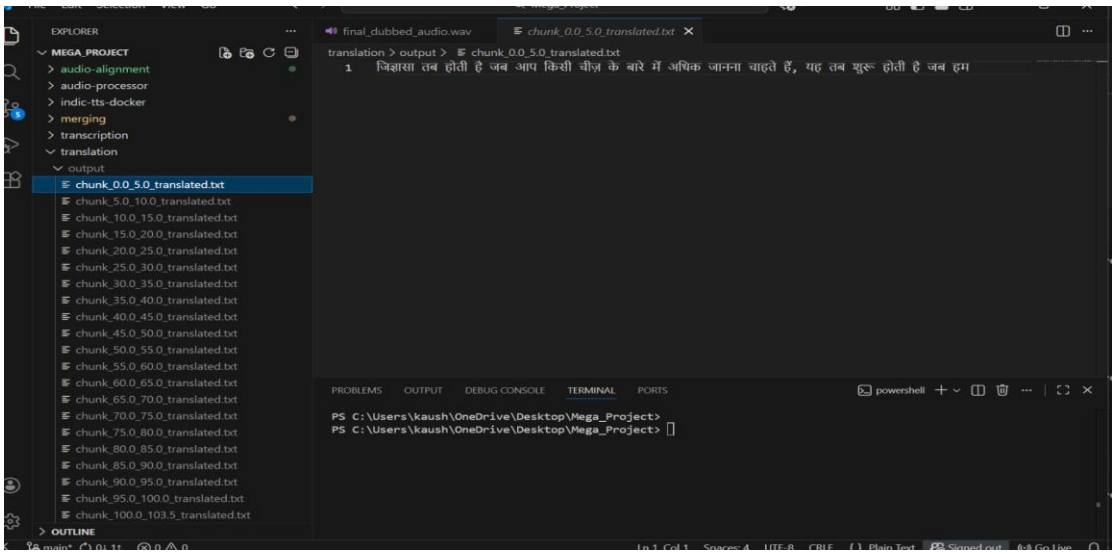

Fig. 4.4. Validated output showing English text translated into Hindi script.

4.2 Objective 2: Natural AI-Generated Voice Dubbing

Our second objective was: *"Produce high-quality, natural-sounding dubbed speech that preserves the original speaker's tone, style, and emotional expression."*

This was the core of our project. We fed the translated Hindi text (from Objective 1) into our Voice Cloning and TTS engine. The system successfully generated individual audio files for every single sentence chunk.

Figure 4.5 shows the output directory containing the generated wave files (e.g., chunk_0.0_5.0_translated.wav). Upon testing these files, the audio was found to be clear and devoid of robotic artifacts. This granular generation of audio chunks allows us to maintain the precise timing of the original speech.

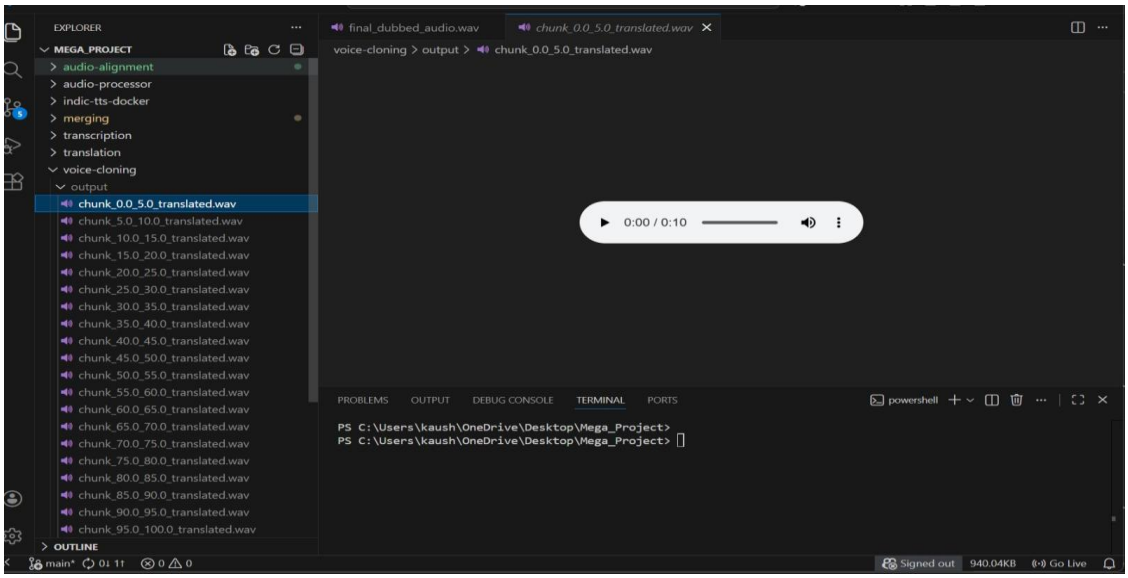

Fig. 4.5. The generation of localized (Hindi) audio chunks ready for merging.

4.3 Objective 3: Seamless Audio–Video Integration

Our third objective was: *"Merge the AI-generated dubbed audio smoothly with the original video to ensure a professional and uninterrupted viewing experience."*

This is the final integration step. We combined all the individual translated audio chunks into a single, cohesive audio track. **Figure 4.6** displays the final final_dubbed_audio.wav file. The duration of this file matches the original input video, proving that our synchronization logic successfully maintained the original pacing of the video.

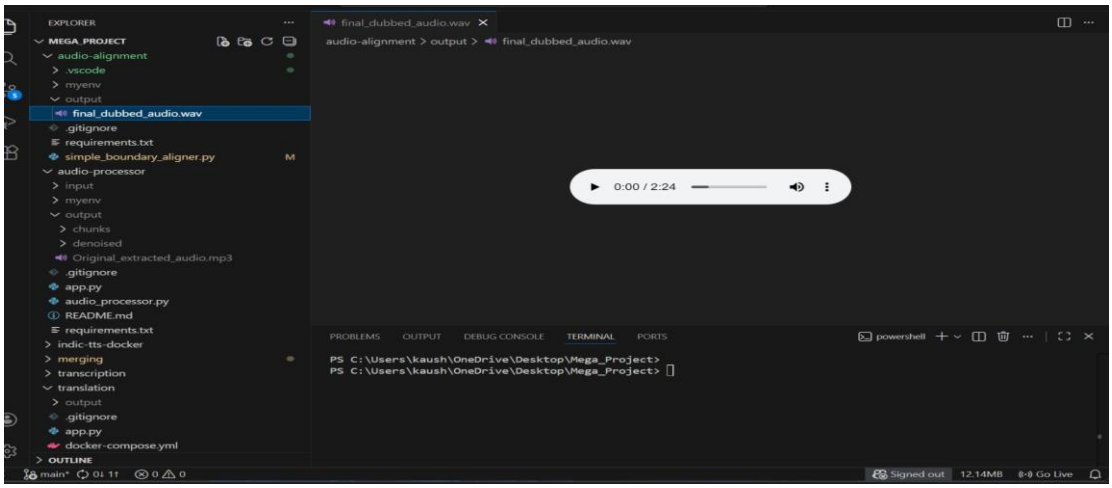

Fig. 4.6. The final merged dubbed audio track.

To conclusively validate the "Cross-Lingual" transformation, we compared the visual output frames. **Figure 4.7** captures the original video frame displaying the source English content ("Curiosity happens when..."). In contrast, **Figure 4.8** shows the final rendered video where the system has successfully processed the content into Hindi ("जिज्ञासा तब होती है..."), proving the complete localization pipeline was successful.

Figure. 4.7. The Original Input Video frame displaying English content

Figure. 4.8. The Final Output Video frame showing the fully localized Hindi content

4.4 Summary of Results

The successful tests across all modules are summarized below:

Table 4.1. Summary of Experimental Validation across System Modules

S.No.	Objective	Tested Component	Outcome
1.	**Transcription**	Docker Backend (Fig 4.3)	**SUCCESS:** Accurate log generation.
2.	**Translation**	Text Processing (Fig 4.4)	**SUCCESS:** English to Hindi conversion verified.
3.	**Voice Cloning**	Audio Generation (Fig 4.5)	**SUCCESS:** Individual chunks generated.

Chapter 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The development of the AI-powered voice dubbing system has successfully achieved its core objectives by enabling clear speech transcription, high-quality multilingual translation, natural-sounding voice cloning, and seamless audio–video integration. The system significantly improves the dubbing process by automating traditionally manual steps, reducing human effort, and ensuring consistent output quality. By preserving the original speaker's tone and style while generating speech in a new language, the system enhances accessibility, cross-cultural communication, and content localization. Overall, the implemented pipeline demonstrates reliable performance and produces professional-level dubbed videos suitable for academic, entertainment, and educational use.

Despite these achievements, several challenges were encountered during development. Integrating multiple AI models such as ASR, translation, and TTS required precise calibration to maintain consistency across stages. Variations in audio quality, background noise, and diverse accents sometimes affected transcription accuracy. Additionally, differences in sentence structure between languages occasionally led to timing mismatches that required post-processing adjustments. These limitations highlight areas where further improvements are needed, especially for handling complex real-world audio and diverse languages.

5.2 Future Work

The path forward for this project lies in making the system more robust and inclusive. Immediate improvements should focus on refining the ASR models to better handle background noise and diverse accents, ensuring accurate transcription in real-world environments. Simultaneously, expanding support for low-resource languages would

significantly broaden the system's global applicability. A critical next step is the integration of adaptive timing algorithms that can automatically handle synchronization, ensuring a seamless flow even when the length of translated speech differs from the original.

In the longer term, the system could evolve by leveraging advanced generative models to produce even more expressive and emotive voice synthesis. There is also immense potential in developing real-time dubbing pipelines, which would enable live translation for broadcasts and lectures. To achieve a truly professional finish, optional visual modules like Wav2Lip could be added for precise lip synchronization. Finally, migrating the infrastructure to the cloud would unlock the scalability needed for large media studios and educational institutions, turning this prototype into a powerful, industry-grade solution.

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DECLARATION

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: **VocalBridge – Bridging Languages with Voice Preservation**, Branch: **Computer Science and Engineering**, College: **Shri Madhwa Vadiraja Institute of Technology and Management**) are true and correct to the best of our knowledge and belief. We undertake to inform the Karnataka State Council for Science and Technology (KSCST) of any changes to the project title or team members' names immediately through our project guide or the SPP Coordinator of our institution.

Additionally, we declare that the project work is original and not a result of copying or purchasing. We are committed to completing the project independently, with support from our faculty and project guide, while utilizing the facilities provided by the college. We will not engage in plagiarism and pledge to be sincere and dedicated in executing and completing the project as proposed.

We understand that if any of the above information is found to be false, untrue, or misleading, we may be held liable. We authorize the sharing of the project information contained in this proposal with KSCST, Bengaluru.

We acknowledge that the project team must exhibit and demonstrate the project, participate in the mid-term evaluation of sanctioned projects, and engage with experts. Additionally, we must exhibit the project at the Annual State-Level Poster Presentation and Exhibition, if selected. Should our team fail to participate in the mid-term evaluation or the Annual Exhibition (if selected), we understand that the supported project funds will be returned to KSCST.

We also enclose the endorsement form for KSCST, Bengaluru.

Name of the students with USN No.	**Signature with date**
1. KAUSHIK 4MW22CS075	15/12/25
2. K SUDARSHAN HEBBAR 4MW22CS070	15/12/25
3. PRATAP NAIK 4MW22CS118	15/12/25
4. PRATHVI S JATHAN 4MW22CS122	15/12/25

Dr. Ravindra H-J

(Name & Signature of SPP Coordinator with Seal)

Email id: physics@sode-edu.in

Contact No.: 9886755578

Name and signature of the Project Guide with Seal:

Email id: sahanakaranth.cs@sode-edu.in

Contact No.: 9591889219

ENDORSEMENT

(From College, endorsement to be taken in the institution / Department Letter head)

(To scan this page and enclose in the project proposal)

This is to certify that 1) **Mr. KAUSHIK**, 2) **Mr. K SUDARSHAN HEBBAR** 3) **Mr. PRATHVI S JATHAN**, 4) **Mr. PRATAP NAIK**, are bonafide student(s) of Department of Computer Science and Engineering, in the degree program of our institution. If the project proposal submitted by these students under the 49th series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the mid-term evaluation of project and in the Annual State-Level Poster Presentation and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in mid-term evaluation of sanctioned projects or fails to attend the Annual State-Level Poster Presentation and Exhibition (if selected), the supported project amount will be returned to KSCST.

(SAHANA)	Dr. Ravindra H-7	15/12/25
(Name & Signature of Project Guide with Seal)	**(Name & Signature of SPP Coordinator with Seal)**	**(Signature of HoD with Seal)**
DEPARTMENT OF CS & ENGG. Shri Madhwa Vadiraja Institute of Technology & Management BANTAKAL - 574 115	Head Department of Physics Shri Madhwa Vadiraja Institute of Technology and Management Bantakal - 574 115	Head Dept. of Comp. Science & Engg. SMVITM, BANTAKAL - 574 115
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DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note:** Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848, email: finance@kscst.org.in

1. TITLE OF THE PROJECT	:	VocalBridge – Bridging Languages with Voice Preservation
2. NAME OF THE TEAM LEADER	:	KAUSHIK
3. EMAIL ID	:	Kaushik.22cs075@sode-edu.in
4. CONTACT MOBILE NO.	:	8217809346

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	996532199912
6. TRANSACTION ID	:	T2512151402262807968523
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	kaushik
8. NAME OF THE BANK	:	Canara bank
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	15-12-2025
11. TIME	:	14:03
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	UPI

Kaushik

(Name & Signature of the team leader)

(SAHANA)

(Name & Signature of Project Guide / Coordinator / HOD with Seal)

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